

Dbms Week3

Question 1:

```
[mysql> DESC Emp;
```

Field	Type	Null	Key	Default	Extra
eid	int	NO	PRI	NULL	
ename	varchar(50)	NO		NULL	
age	int	YES		NULL	
salary	double	YES		NULL	

```
4 rows in set (0.004 sec)
```

```
[mysql> DESC Dept;
```

Field	Type	Null	Key	Default	Extra
did	int	NO	PRI	NULL	
dname	varchar(50)	YES		NULL	
budget	double	YES		NULL	
managerid	int	YES	MUL	NULL	

```
4 rows in set (0.002 sec)
```

```
[mysql> DESC Works;
```

Field	Type	Null	Key	Default	Extra
eid	int	NO	PRI	NULL	
did	int	NO	PRI	NULL	
pcttime	int	YES		100	

```
3 rows in set (0.003 sec)
```

Question 2:

-- (a)

```
CREATE TABLE Hotel (...);
```

```
CREATE TABLE Room (...);
```

```
CREATE TABLE Guest (...);
```

```
CREATE TABLE Booking (...);
```

-- (b)

```
ALTER TABLE Hotel ALTER COLUMN city SET DEFAULT 'London';
```

```
ALTER TABLE Room MODIFY price REAL NOT NULL;  
ALTER TABLE Guest ADD CONSTRAINT guest_name_unique UNIQUE (guestName);  
ALTER TABLE Room ADD CONSTRAINT room_price_chk CHECK (price > 0);
```

```
-- (c)  
CREATE TABLE Tourism (...);
```

```
-- (d)  
SHOW CREATE TABLE Tourism;  
ALTER TABLE Tourism DROP FOREIGN KEY ...;  
ALTER TABLE Hotel MODIFY hotelName VARCHAR(50);  
ALTER TABLE Tourism RENAME TO Tour_Information;  
ALTER TABLE Tour_Information RENAME COLUMN guestNo TO Guest_Num;  
ALTER TABLE Hotel ADD Rating INT;  
ALTER TABLE Hotel DROP COLUMN Rating;
```

```
-- (e)  
SHOW CREATE TABLE Booking;  
ALTER TABLE Booking DROP FOREIGN KEY ...;  
ALTER TABLE Room DROP PRIMARY KEY;  
ALTER TABLE Room ADD PRIMARY KEY (roomNo, type);
```

```
-- (f)  
SHOW TABLES;  
TRUNCATE TABLE Tour_Information;  
SHOW TABLES;  
DROP TABLE Tour_Information;
```

```
SHOW TABLES;
```

```
| Tables_in_lab3dbms |
+-----+
| booking            |
| Dept              |
| Emp               |
| guest             |
| hotel             |
| room              |
| tourism_info       |
| Works             |
+-----+
8 rows in set (0.002 sec)
```

```
[mysql> truncate table tourism_info;
Query OK, 0 rows affected (0.011 sec)
```

```
[mysql> show tables;
+-----+
| Tables_in_lab3dbms |
+-----+
| booking            |
| Dept              |
| Emp               |
| guest             |
| hotel             |
| room              |
| tourism_info       |
| Works             |
+-----+
8 rows in set (0.003 sec)
```

```
[mysql> SELECT * FROM Tour_Information;
ERROR 1146 (42S02): Table 'lab3dbms.tour_information' doesn't exist
[mysql> DROP TABLE Tour_Information;
ERROR 1051 (42S02): Unknown table 'lab3dbms.tour_information'
[mysql> SELECT * FROM tourism_info;
Empty set (0.002 sec)
```

```
[mysql> drop table tourism_info;
Query OK, 0 rows affected (0.007 sec)
```

```
[mysql> show tables;
+-----+
| Tables_in_lab3dbms |
+-----+
| booking            |
| Dept              |
| Emp               |
| guest             |
| hotel             |
| room              |
| Works             |
+-----+
7 rows in set (0.003 sec)
```

Question 3 👍

```
-- =====  
-- COMPANY DATABASE - COMPLETE DDL  
-- =====
```

```
-- 1. EMPLOYEE  
-- SUPERSSN is a self-referencing foreign key
```

```
CREATE TABLE EMPLOYEE (  
    FNAME VARCHAR(20) NOT NULL,  
    MINIT CHAR(1),  
    LNAME VARCHAR(20) NOT NULL,  
    SSN CHAR(9) PRIMARY KEY,  
    BDATE DATE,  
    ADDRESS VARCHAR(100),  
    SEX CHAR(1),  
    SALARY DECIMAL(10,2),  
    SUPERSSN CHAR(9),  
    DNO INT,  
  
    FOREIGN KEY (SUPERSSN)  
        REFERENCES EMPLOYEE(SSN)  
);
```

```
-- 2. DEPARTMENT  
-- MGRSSN references EMPLOYEE
```

```
CREATE TABLE DEPARTMENT (  
    DNAME VARCHAR(30) NOT NULL,  
    DNUMBER INT PRIMARY KEY,  
    MGRSSN CHAR(9),  
    MGRSTARTDATE DATE,  
  
    FOREIGN KEY (MGRSSN)  
        REFERENCES EMPLOYEE(SSN)  
);
```

```
-- 3. Add EMPLOYEE.DNO foreign key  
-- DNO references DEPARTMENT.DNUMBER
```

```
ALTER TABLE EMPLOYEE
```

```
ADD CONSTRAINT fk_employee_department
FOREIGN KEY (DNO)
REFERENCES DEPARTMENT(DNUMBER);
```

```
-- 4. DEPT_LOCATIONS
-- Composite primary key: (DNUMBER, DLOCATION)
```

```
CREATE TABLE DEPT_LOCATIONS (
  DNUMBER INT,
  DLOCATION VARCHAR(30),

  PRIMARY KEY (DNUMBER, DLOCATION),

  FOREIGN KEY (DNUMBER)
    REFERENCES DEPARTMENT(DNUMBER)
);
```

```
-- 5. PROJECT
-- PNUMBER is primary key
-- DNUM references DEPARTMENT
```

```
CREATE TABLE PROJECT (
  PNAME VARCHAR(30),
  PNUMBER INT PRIMARY KEY,
  PLOCATION VARCHAR(30),
  DNUM INT,

  FOREIGN KEY (DNUM)
    REFERENCES DEPARTMENT(DNUMBER)
);
```

```
-- 6. WORKS_ON
-- Composite primary key: (ESSN, PNO)
-- Both foreign keys use CASCADE
```

```
CREATE TABLE WORKS_ON (
  ESSN CHAR(9),
  PNO INT,
  HOURS DECIMAL(5,2),

  PRIMARY KEY (ESSN, PNO),
```

```

FOREIGN KEY (ESSN)
  REFERENCES EMPLOYEE(SSN)
  ON DELETE CASCADE
  ON UPDATE CASCADE,

FOREIGN KEY (PNO)
  REFERENCES PROJECT(PNUMBER)
  ON DELETE CASCADE
  ON UPDATE CASCADE
);

-- 7. DEPENDENT
-- Composite primary key: (ESSN, DEPENDENT_NAME)

```

```

CREATE TABLE DEPENDENT (
  ESSN CHAR(9),
  DEPENDENT_NAME VARCHAR(30),
  SEX CHAR(1),
  BDATE DATE,
  RELATIONSHIP VARCHAR(30),

  PRIMARY KEY (ESSN, DEPENDENT_NAME),

  FOREIGN KEY (ESSN)
    REFERENCES EMPLOYEE(SSN)
);

```

```

-- =====
-- PART (d)
-- Add CHECK constraint on SALARY
-- Set default value of 1 for DNO
-- =====

```

```

ALTER TABLE EMPLOYEE
ADD CONSTRAINT employee_salary_chk
CHECK (SALARY > 0);

```

```

ALTER TABLE EMPLOYEE
ALTER COLUMN DNO SET DEFAULT 1;

```

```
-- =====  
-- PART (e)  
-- Display structure of every table  
-- =====
```

```
DESC EMPLOYEE;
```

```
DESC DEPARTMENT;
```

```
DESC DEPT_LOCATIONS;
```

```
DESC PROJECT;
```

```
DESC WORKS_ON;
```

```
DESC DEPENDENT;
```



```
mysql> DESC EMPLOYEE;
```

Field	Type	Null	Key	Default	Extra
FNAME	varchar(20)	YES		NULL	
MINIT	char(1)	YES		NULL	
LNAME	varchar(20)	YES		NULL	
SSN	char(9)	NO	PRI	NULL	
BDATE	date	YES		NULL	
ADDRESS	varchar(100)	YES		NULL	
SEX	char(1)	YES		NULL	
SALARY	decimal(10,2)	YES		NULL	
SUPERSSN	char(9)	YES		NULL	
DNO	int	YES	MUL	NULL	

```
10 rows in set (0.004 sec)
```

```
mysql> DESC DEPARTMENT;
```

Field	Type	Null	Key	Default	Extra
DNAME	varchar(30)	YES		NULL	
DNUMBER	int	NO	PRI	NULL	
MGRSSN	char(9)	YES	MUL	NULL	
MGRSTARTDATE	date	YES		NULL	

```
4 rows in set (0.002 sec)
```

```
mysql> DESC DEPT_LOCATIONS;
```

Field	Type	Null	Key	Default	Extra
DNUMBER	int	NO	PRI	NULL	
DLOCATION	varchar(30)	NO	PRI	NULL	

```
2 rows in set (0.001 sec)
```

```
mysql> DESC PROJECT;
```

Field	Type	Null	Key	Default	Extra
PNAME	varchar(30)	YES		NULL	
PNUMBER	int	NO	PRI	NULL	
PLOCATION	varchar(30)	YES		NULL	
DNUM	int	YES	MUL	NULL	

```
4 rows in set (0.001 sec)
```